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UML Class Diagram Modelling Assignment

ONLINE SHOPPING MANAGEMENT SYSTEM

1. Introduction

The Online Shopping Management System is a software system designed to manage the main activities involved in online shopping. It helps customers browse products, manage their shopping carts, place orders, and record payments.

The purpose of this project is to model the structure of the system using a UML class diagram. The diagram identifies the main classes, their attributes, their methods, and the relationships between them.

2. Business Idea

The business idea is to develop an online shopping platform where customers can select products, add them to their shopping carts, place orders, and make payments.

The system also allows the management of product information, stock quantities, customer information, order details, and payment records.

3. Project Scope

Included features:

- Customer registration and profile management.
- Product information and availability management.
- Shopping cart management.
- Adding and removing products from the cart.
- Order creation and tracking.
- Payment information management.

Excluded features:

- Physical product delivery management.
- Supplier management.
- Warehouse management.
- Advanced financial reporting.
- Development of a complete, functional website or mobile application.

4. Identified Classes

The system contains seven main classes.

4.1 Customer

Represents a person who uses the platform to purchase products. It stores customer information and provides methods for registration, profile updates, and order viewing.

4.2 Product

Represents a product available for purchase. It stores the product name, description, price, and stock quantity. Its methods support price updates, stock updates, and availability checks.

4.3 ShoppingCart

Represents a customer's active shopping cart. It allows products to be added or removed and calculates the total price of the selected items.

4.4 CartItem

Represents an individual product entry in a shopping cart. It stores the quantity and unit price and calculates the subtotal.

4.5 Order

Represents a confirmed purchase made by a customer. It stores the order identifier, date, status, and total amount. It supports order placement, cancellation, and total calculation.

4.6 OrderItem

Represents an individual product entry within an order. It stores the quantity and unit price and calculates the subtotal for that item.

4.7 Payment

Represents a payment record associated with an order. It stores the payment amount, date, method, and status. It provides methods for processing and verifying payments.

5. Relationships Between Classes

The UML class diagram contains seven relationships.

5.1 Customer — ShoppingCart

This is an association. A customer can have zero or one active shopping cart, and each shopping cart belongs to exactly one customer.

5.2 Customer — Order

This is an association. A customer can place multiple orders, while each order belongs to exactly one customer.

5.3 ShoppingCart — CartItem

This is a composition. A shopping cart contains zero or more cart items. The cart items belong to that cart and are managed as part of it.

5.4 CartItem — Product

This is an association. Each cart item refers to exactly one product, while a product can appear in multiple cart items.

5.5 Order — OrderItem

This is a composition. An order contains one or more order items. These items belong to the order they describe.

5.6 OrderItem — Product

This is an association. Each order item refers to exactly one product, while a product can appear in multiple order items.

5.7 Order — Payment

This is an association. An order can have zero or more payment records, allowing the system to represent an unpaid order or multiple payment attempts.

6. Conclusion

The UML class diagram provides a structural representation of the Online Shopping Management System. It identifies the main entities, their responsibilities, and the relationships required to manage customers, products, shopping carts, orders, and payments.